



Software for  
Theoretical  
Chemical  
Kinetics and  
IRC Analysis

# Eyringpy

version 3.0

**USER MANUAL**

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# 1 Getting started

## 1.1 What is *Eyringpy*?

*Eyringpy* is a user-friendly program, launched in 2019, for computing rate constants with the transition state theory (TST),<sup>1,2</sup> in the gas phase and in solution. In particular, the Gibbs activation energy is obtained by computing from scratch the canonical partition functions.<sup>3</sup> Unimolecular and bimolecular reactions with one or two products are supported. Rate constants of bimolecular reactions involving the formation of pre-reactive and/or product complexes are also estimated.<sup>4</sup> The reaction symmetry is considered.<sup>5</sup> One dimensional Wigner<sup>6</sup> and Eckart<sup>7</sup> tunneling corrections are also available. To compute rate constants in solution, *Eyringpy* employs the Collins–Kimball theory to include the diffusion-limit,<sup>8-12</sup> the Marcus theory for electron transfer (ET) processes,<sup>13-16</sup> and the molar fractions to account for the effect of pH for aqueous reactions.<sup>17</sup> Now, in *Eyringpy* 3.0 we included the estimation of rate constants through the variational transition state theory (CVT)<sup>18</sup> and the hindered rotor models of Pitzer and Gwinn,<sup>19</sup> and Ayala and Schlegel.<sup>20</sup>

*Eyringpy* can also scan the evolution of primitive changes along the intrinsic reaction coordinate (IRC).<sup>21, 22</sup> The primitive changes that can be studied in this release are the bond distances, angles, Wiberg bond indices,<sup>23</sup> natural charges, dipole moment, orbital energies (and related properties), and reaction forces.<sup>24</sup> A spline interpolator from *Python*'s Scipy library is used to smooth the energy profiles. In addition, a discriminator to remove points causing noise in the energy profiles is also available. The *Python*'s Matplotlib library is used for plotting the evolution of the primitive changes along the IRC.

## 1.2 How to obtain *Eyringpy*?

Go to the Merino Research Group's webpage (<https://www.theochemmerida.org/eyringpy>), fill out the *Eyringpy* order form online, and download the *Eyringpy* version for the required operating system.



### 1.3 Software requirements

*Eyringpy* is a command-line program written in *Python3*. It is compatible with Windows, Linux, and Mac OS, and is distributed as a stand-alone executable. *Eyringpy* requires the output files of *Gaussian* (09 or 16) (\*.out or \*.log) with the frequency analysis of the species involved in the reaction (reactant(s), product(s), transition state, and pre-reactive and/or product complexes, if necessary), and/or with the IRC data. Besides, the *Eyringpy*'s input file (foldername.eif) is also needed. It is not mandatory to install *Eyringpy* in the same computer where *Gaussian* is running. For more information about *Gaussian*, visit its web page: <https://www.gaussian.com>.

### 1.4 *Eyringpy* software content

*Eyringpy* is distributed in a compressed (\*.zip for Windows or \*.tar.gz for Linux and Mac OS) file containing the directory eyringpyversion\_operatingsystem\_architecture. This directory has the following files and subdirectories:

**bin:** Contains the *Eyringpy* binary.

**manual:** Contains the *Eyringpy* manual.

**test:** Contains the *Eyringpy* tests.

**Note:** The eyringpy binary is named eyringpy.exe in Windows and eyringpy in Linux and Mac OS.

### 1.5 Installing *Eyringpy*

The installation of *Eyringpy* is simple and straightforward. However, if you have problems installing it, contact us for assistance at [eyringpy2018@gmail.com](mailto:eyringpy2018@gmail.com). To install *Eyringpy* you have to go through the steps for the operating system you chose.

#### 1.5.1 Windows

1. Download the eyringpyversion\_windows\_xbits.zip file, move it to the directory of your choice, and uncompress it.



2. Open the command prompt (cmd or cmd.exe):

- \* Press Win+R to open the run box, type cmd, and press Enter.

**Note:** For detailed information about the Windows commands, please visit: <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/windows-commands>.

3. Go to the eyringpyversion\_windows\_xbits directory:

```
> cd path-to-eyringpyversion_windows_xbits-directory
```

**Example:**

```
> cd Documents\Programs\eyringpy3.0_windows_64bits
```

4. Type `dir /s` to display all the files and subdirectories in the eyringpyversion\_windows\_xbits directory and make sure it has the content shown in **Section 1.4**.

5. Add an eyringpy alias (e.g. eyringpy) to a Batch script to run the program in any directory:

- \* Open notepad (by the graphical user interface (GUI) or type notepad in the cmd and press Enter) and paste the lines shown in Figure 1.

```
@echo off
title command prompt
doskey eyringpy=path-to-eyringpyversion_windows_xbits-directory\bin\eyringpy.exe $1
```

**Figure 1.** Batch script (profile.bat) to generate the eyringpy alias in Windows.

**Example:**

```
doskey eyringpy=C:\Users\Eugenia\Documents\Programs
\eyringpy3.0_windows_64bits\bin\eyringpy.exe $1
```

- \* Save the file as profile.bat in the Desktop or in Documents.

**Note:** To edit the profile.bat file, right-click on its icon and select **Edit**.

- \* Create a desktop shortcut of the cmd.
- \* Right-click the cmd shortcut, select **Properties** from the menu, and click the **Shortcut** tab.



- \* In the Target field, type `path-to-cmd.exe /k path-to-profile.bat` and click Ok.

**Example:**

Target: `%windir%\system32\cmd.exe` /k  
`C:\Users\Eugenia\Desktop\profile.bat`

**Note:** For more information about the doskey command and Batch scripts please visit:

<https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/doskey> and

<https://docs.microsoft.com/en-us/azure/devops/pipelines/tasks/utility/batch-script?view=azure-devops>, respectively.

6. Run a test job of *Eyringpy*:

- \* Double-click the cmd Shortcut.
- \* Go to one of the test' subdirectories.

**Example:**

```
> cd Documents\Programs\eyringpy3.0_windows_64bits\test\kinetics\gas\g09\ch4_cl
```

- \* Type the `eyringpy` alias. If the installation is successful, the *Eyringpy* welcome-help message (Figure 2) will be displayed.



```
Eyringpy version 3.0
Authors: Eugenia Dzib, Alan Quintal, and Gabriel Merino
Cinvestav Merida, 2025

Modules

* Kinetics computes rate constants in the gas phase and in solution

To run this module: eyringpy -kin

* IRC-analysis monitors the evolution of primitive changes along the IRC

To run this module: eyringpy -irc

Please check the parameters in the input file foldername.eif
```

**Figure 2.** *Eyringpy* welcome-help message.

**Note:** The *Eyringpy* welcome-help message in Figure 2 is displayed when there is an *Eyringpy* input file in the current working directory. Otherwise, the last line in the welcome-help message is replaced by “If an input file is required, please run *eyringpy -kin* or *eyringpy -irc* as the case may be.”

- \* Type *eyringpy -kin* or *eyringpy -irc* depending on the type of computation you choose.
- \* Type *dir* to list the directory contents and make sure the *Eyringpy* output file *foldername.eof* is present.
- \* Open the output file *foldername.eof* (by the GUI or type *foldername.eof* in the cmd and press Enter). If everything is OK, the “Eyringpy terminated normally” legend is written at the end of the file.

### 1.5.2 Linux and Mac OS

1. Download the *eyringpyversion\_linux\_or\_macos\_xbits.tar.gz* file.
2. Open the terminal:
  - \* Press Ctrl+Alt+T (Linux)



- \* Press Command+Space to launch Spotlight and type terminal.app (Mac OS)
- 3. Move the `eyringpyversion_linux_or_macos_xbits.tar.gz` file to the directory of your choice and uncompress it:  

```
$ mv eyringpyversion_linux_or_macos_xbits.tar.gz destination_directory
```

```
$ cd destination_directory
```

```
$ tar -xzf eyringpyversion_linux_or_macos_xbits.tar.gz
```
- 4. Go to the `eyringpyversion_linux_or_macos_xbits` directory:  

```
$ cd path-to-eyringpyversion_linux_or_macos_xbits-directory
```
- 5. Type `ls -R` to display all the files and subdirectories in the `eyringpyversion_linux_or_macos_xbits` directory and make sure it has the content shown in **Section 1.4**.
- 6. Add the *Eyringpy* path to the PATH environment variable to run *Eyringpy* in any directory.
  - \* For Linux (bash/sh shell):
    - Open the `.bashrc` file with a text editor (*e.g.* `vi` or `emacs`):  

```
$ vi ~/.bashrc
```
    - Add `export PATH="path-to-bin-directory:$PATH"` to the file and save it.
    - Source the `.bashrc` file  

```
$ source ~/.bashrc
```
  - \* For Linux (csh/tcsh shell):
    - Open the `.cshrc` file with a text editor (*e.g.* `vi` or `emacs`):  

```
$ vi ~/.cshrc
```
    - Add `setenv PATH $PATH:"path-to-bin-directory"` to the file and save it.
    - Source the `.cshrc` file  

```
$ source ~/.cshrc
```
  - \* For Mac OS:
    - Open the `.bash_profile` file with a text editor (*e.g.* `vi` or `emacs`):  

```
$ vi ~/.bash_profile
```





- Add `export PATH="path-to-bin-directory:$PATH"` to the file and save it.
- Source the `.bash_profile` file  
`$ source ~/.bash_profile`

7. Run a test job of *Eyringpy*:

- \* Go to one of the `test`' subdirectories.

**Example:**

```
$ cd
Programs/eyringpyversion_linux_or_macos_xbits/test/kinetics/gas/
g09/ch4_cl
```

- \* Type `eyringpy`. If the installation is successful, the *Eyringpy* welcome-help message (Figure 2) will be displayed.
- \* Type `eyringpy -kin` or `eyringpy -irc` depending on the kind of computation you choose.
- \* Type `ls` to list the directory content and make sure the *Eyringpy* output file `foldername.eof` is present.
- \* Open the output file `foldername.eof` (by the GUI or type `foldername.eof` in the cmd and press Enter). If everything is OK, the "Eyringpy terminated normally" legend is written at the end of the file.

**Note:** Do not perform *Eyringpy* computations in the `bin` directory.

## 1.6 *Eyringpy* tests

The *Eyringpy* package contains a series of sample tests. They have been designed to be short and simple and serve mainly to illustrate the use of the input keywords, necessary files, and type of results regarding the different computations offered by *Eyringpy*. The descriptions are brief and the detailed information about how to use the different input keywords is provided in **Section 2**. The necessary files for the sample tests are located in subdirectories of the `eyringpyversion_operatingsystem_architecture/test` directory, for instance, the `eyringpyversion_operatingsystem_architecture/test/kinetics/tst/gas/g09/ch4cl`



directory contains the kinetics of the reaction  $\text{CH}_4 + \cdot\text{Cl}$  in the gas phase employing the electronic structure results of *Gaussian 09*. Each sample test subdirectory contains the *Eyringpy* input file (foldername.eif), and input (\*.inp) and output (\*.out) files of *Gaussian (09 or 16)*. The terminology used in the *Eyringpy* output file(s) is described in **Section 3**.

### 1.6.1 Rate constants

The different functionalities of *Eyringpy* for computing rate constants are exemplified using two reaction systems: a) The hydrogen abstraction from  $\text{CH}_4$  by  $\cdot\text{Cl}$  in the gas phase shows how to include the tunneling effect and b) the hydrogen transfer from  $\text{NH}_3$  by  $\cdot\text{OH}$  illustrates the use of the available effects for reactions in solution (tunneling, diffusion, electron transfer, and pH).

#### 1.6.1.1 Hydrogen abstraction from $\text{CH}_4$ by $\cdot\text{Cl}$

All geometries were optimized at the  $\text{MP2}^{25}/6\text{-}311\text{G}(3\text{d},2\text{p})^{26}$  level. The nature of all the stationary points was determined by performing a frequency analysis at the same level. The intrinsic reaction coordinate (IRC) (<https://gaussian.com/irc/>) was computed to ensure the transition state connects reactants and products. All computations were done with *Gaussian 09*<sup>27</sup> and *Gaussian 16*.<sup>28</sup> The rate constants were computed using the transition state theory in combination with the Eckart tunneling (TST/ECK) in the gas phase within the temperature range 200-300 K. All the necessary files for the test are stored in the `eyringpyversion_operatingsystem_architecture/test/kinetics/tst/gas/g09/ch4cl` directory. The *Eyringpy* input file (ch4cl.eif) for this computation is depicted in Figure 3.

|        |              |
|--------|--------------|
| METHOD | TST          |
| TUNNEL | ECK          |
| TEMP   | 200 300 10   |
| REACT1 | ch4.out      |
| REACT2 | cl.out       |
| TS     | ch4cl_ts.out |
| PROD1  | ch3.out      |
| PROD2  | hcl.out      |
| RXSYM  | 4            |



**Figure 3.** *Eyringpy* input file (ch4cl.eif) of the hydrogen abstraction from CH<sub>4</sub> by  $\cdot\text{Cl}$ .

### 1.6.1.2 Hydrogen transfer from NH<sub>3</sub> by $\cdot\text{OH}$

All geometries were optimized at the M06-2X<sup>29</sup>/6-311++G(d,p) level. The nature of all the stationary points was determined by performing a frequency analysis. The intrinsic reaction coordinate (IRC) computation was carried out to verify the transition state connects reactants and products. Solvent effects were mimicked with the solvation model based on density (SMD)<sup>30</sup> using water as solvent. All computations were done with *Gaussian 09*<sup>27</sup> and *Gaussian 16*.<sup>28</sup> The rate constants were computed according to the transition state theory combined with the Eckart tunneling (TST/ECK) at the 1M standard state, for the temperature range 300-350 K. In addition, the diffusion-limit was included by calculating the apparent rate constant using the Collins-Kimball theory. To introduce this effect, the reaction distance (*i.e.*, the distance required for a bimolecular reaction to take place) and radii of the solutes (NH<sub>3</sub> and  $\cdot\text{OH}$ ) are needed. The reaction distance was obtained by the method of Galano and Alvarez-Idaboy.<sup>17</sup> For hydrogen transfer reactions, it consists in taking, from the transition state, the distance between the two atoms involved in the H abstraction as the reaction distance (in this case, the N-O distance). The radii of the solutes were computed with *Gaussian 09* (<https://gaussian.com/volume/>). The influence of pH was considered by calculating the molar fractions of the NH<sub>4</sub><sup>+</sup>/NH<sub>3</sub> acid-base pair.<sup>17</sup> The pK<sub>a</sub> of the ammonium cation was obtained from experimental results<sup>31, 32</sup> found in the literature, but for cases in which there are no such experimental evidence, we recommend obtaining theoretical pK<sub>a</sub>(s) using the protocol of Galano and co-workers.<sup>33-36</sup> The tests files are located in the `eyringpyversion_operatingsystem_architecture/test/kinetics/tst/solution/gaussianversion/nh3oh` directory. The *Eyringpy* input file (nh3oh.eif) for this computation is shown in Figure 4.



|         |              |
|---------|--------------|
| METHOD  | TST          |
| TUNNEL  | ECK          |
| DIFF    | true         |
| AB      | 0            |
| PHASE   | sol          |
| TEMP    | 300 350 10   |
| STEMP   | 298.15       |
| REACT1  | nh3.out      |
| REACT2  | oh.out       |
| TS      | nh3oh_ts.out |
| PROD1   | nh2.out      |
| PROD2   | h2o.out      |
| RXSYM   | 3            |
| SOLVENT | water        |
| RXD     | 2.4          |
| RAD1    | 2.63         |
| RAD2    | 2.64         |
| PH      | 0 14 1       |
| PKA_R1  | 9.25         |

**Figure 4.** *Eyringpy* input file (nh3oh.eif) of the hydrogen abstraction from  $\text{NH}_3$  by  $\cdot\text{OH}$ .

### 1.6.1.3 Hydrogen abstraction from formic acid by $\cdot\text{OH}$

Geometry optimizations and frequency analysis were carried out at the M05-2X<sup>37</sup>/6 311++G(d,p) level using *Gaussian 09* Rev D.01.<sup>38</sup> Local minima and TSs were identified by their number of imaginary frequencies (0 and 1, respectively). Unrestricted calculations were used for open-shell systems. To confirm the TSs connect reactants and products, the IRC<sup>21, 22</sup> was computed using the local quadratic approximation (LQA)<sup>39, 40</sup> for the predictor step. The MEP was traced by following the IRC downhill from the TS to the reactants and products. Mass- weighted internal coordinates were used without reorientation or symmetry adjustments. For each TS, one hundred points were computed on both sides of the path with a step size of 0.05 amu<sup>1/2</sup> bohr. The rate constants were computed using the variational transition state theory in combination with the Eckart tunneling (CVT/ECK) in the gas phase within the temperature range 296-440 K. All the necessary files for the test are stored in the



eyringpyversion\_operatingsystem\_architecture/test/kinetics/cvt/gas/g09/hcooh\_oh directory. The *Eyringpy* input file (cvt1a.eif) for this computation is depicted in Figure 5.

|            |            |
|------------|------------|
| METHOD     | CVT        |
| TEMP       | 300 500 20 |
| REACT1     | hcooh.out  |
| REACT2     | oh.out     |
| TS         | ts1a.out   |
| PROD1      | hcoo.out   |
| PROD2      | h2o.out    |
| RXSYM      | 1          |
| IRC        | 1a_irc.out |
| PATHPOINTS | ircp       |

**Figure 5.** *Eyringpy* input file (hcooh\_oh.eif) of the hydrogen abstraction from HCOOH by  $\cdot\text{OH}$ .

#### 1.6.1.4 Hydrogen abstraction from methane by $\cdot\text{OH}$

All geometries were fully optimized at the BhandHLYP/6-311G(d,p) level. Harmonic vibrational frequency analysis confirmed the nature of each stationary point. To improve accuracy, single-point energies were calculated at the CCSD(T)/6-311G(d,p) level, with Gibbs free energies incorporating thermal contributions from BhandHLYP/6-311G(d,p). The IRC was performed using the LQA<sup>39, 40</sup> approximation to verify that the transition states connect the correct reactants and products. Parameters for hindered rotors, including torsional frequency, symmetry, and reduced moment of inertia, were determined using Ayala and Schlegel's algorithm.<sup>20</sup> The torsional potential was scanned in 20° increments from 0 to 360°. The hindered rotor analysis was performed with *Gaussian 09* Rev. D.01 (<https://gaussian.com/freq/>). All computations were carried out using *Gaussian 09* Rev. D.01. The rate constants were computed *via* TST. Tunneling factors were computed using a one-dimensional Eckart potential energy barrier.<sup>20</sup> For low-frequency vibrational modes corresponding to internal rotations, *Eyringpy* employed the one-dimensional models of Pitzer-Gwinn<sup>19</sup> and Ayala-Schlegel<sup>20</sup> to evaluate partition functions and thermodynamic properties. All the necessary files for the test are stored in the



eyringpyversion\_operatingsystem\_architecture/test/kinetics/hr/gas/g09/ch4oh directory. The *Eyringpy* input file (ch4oh.eif) for this computation is depicted in Figure 6.

|         |              |
|---------|--------------|
| METHOD  | TST          |
| HINDROT | AS           |
| TEMP    | 300 650 50   |
| REACT1  | ch4.out      |
| REACT2  | oh.out       |
| TS      | ch4oh-ts.out |
| PROD1   | ch3.out      |
| PROD2   | h2o.out      |
| RXSYM   | 12           |

**Figure 6.** *Eyringpy* input file (ch4oh.eif) of the hydrogen abstraction from CH<sub>4</sub> by ·OH.

### 1.6.2 IRC Analysis

In this case, to show the versatility of *Eyringpy* to analyze the evolution of primitive changes along the IRC we chose two reactions. The Cope rearrangement of 1,5-hexadiene illustrates the reaction force analysis and the double group transfer from ethane to ethylene shows the structural and electronic analysis.

All molecular structures were fully optimized at the M11<sup>41</sup>/6-31+G(d,p)<sup>42, 43</sup> level. All minima and TSs structures were verified by harmonic vibrational frequency analysis. The MEP was computed through the IRC in mass-weighted internal coordinates,<sup>44, 45</sup> using a different step size for each test case. All the computations were done with *Gaussian 09*<sup>27</sup> and *Gaussian16*.<sup>28</sup>

#### 1.6.2.1 [3,3] Cope rearrangement of 1,5-hexadiene

The reaction force analysis (RFA) of the degenerate [3,3] Cope rearrangement of 1,5-hexadiene was performed through the theory developed by the group of Toro-Labbé. The IRC was computed using a step size of 0.03 amu<sup>1/2</sup> bohr. The step size is one of the main parameters that must be chosen to obtain reliable IRCs. Its typical value is 0.10 amu<sup>1/2</sup> bohr, but it must be selected based



on the energy profile behavior. Small step sizes are appropriate for curved reaction paths, while large step sizes are better for flat reaction paths. If the step size is not correctly chosen it could be difficult or even impossible to obtain the MEP that connects reactant(s) and product(s). Besides, this can lead to noisy energy profiles and produce deviations in the activation barriers. The `SPLINE` interpolator was used to illustrate how to smooth the energy profile despite in this case is not necessary. Moreover, the discriminator of the IRC-Analysis module is used to remove the points causing noise in the energy profile, conserving the values of the reaction works and activation energies. The test files are located in the `eyringpyversion_operatingsystem_architecture/test/irc/rforce/gaussianversion/c6h10` directory. The *Eyringpy* input file (`c6h10.eif`) for this calculation is given in Figure 5.

|         |               |
|---------|---------------|
| REACT1  | c6h10.out     |
| TS      | c6h10_ts.out  |
| PROD1   | c6h10.out     |
| IRC     | c6h10_irc.out |
|         |               |
| RFORCE  | true          |
| NPOINTS | 5000          |
| DISCRIM | true          |
|         |               |
| XYZ     | true          |
| PLOT    | rforce        |

**Figure 6.** *Eyringpy* input file (`c6h10.eif`) of the [3,3] Cope rearrangement of 1,5-hexadiene.

### 1.6.2.2 Double group transfer from ethane to ethylene

The properties monitored along the MEP of the electrocyclization of double group transfer from ethane to ethylene are the bond length, angle, Wiberg bond index, natural charges, dipole moment, and HOMO-LUMO orbital energies. The population analysis was performed with *Gaussian 09*<sup>27</sup> and *Gaussian16*<sup>28</sup> (<https://gaussian.com/population>). The IRC was computed using a step size of  $0.10 \text{ amu}^{1/2} \text{ bohr}$ . The necessary files for the sample test are stored in the `eyringpyversion_operatingsystem_xbits/test/irc/property/gaussianversion/c2h6`



\_c2h4 directory. The *Eyringpy* input file (c2h6\_c2h4.eif) for this computation is depicted in Figure 6.

|              |                                                |
|--------------|------------------------------------------------|
| IRC          | c2h6_c2h4_irc.out                              |
| ANGLE        | 1-4-9 5-6-10                                   |
| BONDLENGTH   | 1-4 9-4 5-6 10-6                               |
| NBODIRECTORY | nbo_out                                        |
| CHARGE       | 1 5 9 10                                       |
| WIBERG       | 1-4 9-4 5-6 10-6                               |
| DIPOLE       | true                                           |
| HOMOLUMO     | true                                           |
| NBOINPUT     | 8GB M11 6-31+G(d,p) NBOREAD                    |
| PLOT         | bondlength angle charge wiberg dipole homolumo |

**Figure 6.** *Eyringpy* input file (c2h6\_c2h4.eif) of the double group transfer from ethane to ethylene.

### 1.7 How to cite *Eyringpy*?

If you publish results obtained with *Eyringpy* in the scientific literature, please include the references as follows:

\* **General references:**

[1] E. Dzib, A. Quintal, and G. Merino, *Eyringpy* 3.0, Cinvestav Mérida, Yucatán, 2025.

\* **If you compute rate constants also cite:**

[2] E. Dzib, J. L. Cabellos, F. Ortiz-Chi, S. Pan, A. Galano, G. Merino. [Eyringpy: A Program for computing Rate Constants in the Gas Phase and in Solution](#). *Int. J. Quantum Chem.* 2019, 119, e.25686.

[3] E. Dzib, A. Quintal and G. Merino. Enhancing [Eyringpy: Accurate Rate Constants with Canonical Variational Transition State Theory and the Hindered Rotor Model](#). *J. Chem. Theory Comput.* 2024, 20, 22, 9999.

\* **If you perform the analysis of primitive changes along the IRC also cite:**





[3] A. Quintal, E. Dzib, F. Ortiz-Chi, P. Jaque, A. Restrepo, and G. Merino. Automating the IRC Analysis within *Eyringpy*. *Int. J. Quantum Chem.* 2021, e26684.

### 1.8 Contact

For more information about *Eyringpy*, to report bugs, and send questions or suggestions for improving its performance, please contact us via [eyringpy2018@gmail.com](mailto:eyringpy2018@gmail.com). If you want to report bugs or errors, please describe your problem in detail and send us the output files (\*.out or \*.log) with the electronic structure results and the *Eyringpy* input file (\*.eif) you are using.

**M. Sc. Eugenia Dzib**

[eugenia.dzib@cinvestav.mx](mailto:eugenia.dzib@cinvestav.mx)

[eugeniadzib@gmail.com](mailto:eugeniadzib@gmail.com)

**Professor Gabriel Merino**

[gmerino@cinvestav.mx](mailto:gmerino@cinvestav.mx)

[gabriel.merino2@gmail.com](mailto:gabriel.merino2@gmail.com)

**M. Sc. Alan Quintal**

[alan.quintal@cinvestav.mx](mailto:alan.quintal@cinvestav.mx)

[alanquintal22@gmail.com](mailto:alanquintal22@gmail.com)

**Professor Filiberto Ortiz Chi**

[fortiz@conacyt.mx](mailto:fortiz@conacyt.mx)

[fortiz666@gmail.com](mailto:fortiz666@gmail.com)



## 2 The *Eyringpy* input file

The input parameters for *Eyringpy* must be specified in a text file named as the folder in which you need to execute the program (*i.e.*, `foldername.eif`). Most of them have reliable default values, that allows having extremely short input files. Parameters that do not have default values must be included. Three types of parameters are available: general, for computing rate constants, and for the analysis of primitive changes along the IRC.

### 2.1 General

#### **REACT $n$**

- \* **Type:** String
- \* **Description:** Frequency output filename of the  $n$  reactant. Depending on the molecularity of reaction,  $n$  takes the values of 1 or 2. If the electronic energy at a different level (*e.g.*, MP $n$ , CCSD(T)) is required, the single-point (sp) output filename should be put next to the frequency one.
- \* **Format:** REACT1 react1.out  
REACT1 react1.out react1-sp.out

#### **PROD $n$**

- \* **Type:** String
- \* **Description:** Frequency output filename of the  $n$  product, where  $n$  takes the values of 1 or 2. If the electronic energy at a different level (*e.g.*, MP $n$ , CCSD(T)) is required, the single-point (sp) output filename should be put next to the frequency one.
- \* **Format:** PROD1 prod1.out  
PROD1 prod1.out prod1-sp.out



## 2.2 Kinetics

### TS

- \* **Type:** String
- \* **Description:** Frequency output filename of the transition state. It is mandatory to include it for computing rate constants, but not when only the thermochemistry is needed. If the electronic energy at a different level (*e.g.*, MP*n*, CCSD(T)) is required, the single-point (sp) output filename should be put next to the frequency one.
- \* **Format:** TS ts.out  
TS ts.out ts-sp.out

### RC

- \* **Type:** String
- \* **Description:** Frequency output filename of the pre-reactive complex. It should be included for reactions where the pre-reactive complex has a lower Gibbs free energy when compared to the reactant(s). If the electronic energy at a different level (*e.g.* MP*n*, CCSD(T)) is required, the single-point (sp) output filename should be put next to the frequency one.
- \* **Format:** RC rc.out  
RC rc.out rc-sp.out

### PC

- \* **Type:** String
- \* **Description:** Frequency output filename of the product complex. It should be included for reactions where the pre-reactive complex has a lower Gibbs free energy when compared to the product(s). If the electronic energy at a different level (*e.g.* MP*n*, CCSD(T)) is required, the single-point (sp) output filename should be put next to the frequency one.
- \* **Format:** PC pc.out  
PC pc.out pc-sp.out

**PRODV<sub>n</sub>**

- \* **Type:** String
- \* **Description:** Frequency output filename of the  $n$  vertical product, where  $n$  takes the values 1 or 2. It is used only for bimolecular single electron transfer (SET) reactions in solution.
- \* **Format:** PRODV1 prodv1.out

**NSYM\_R<sub>n</sub>**

- \* **Type:** Integer
- \* **Default value:** Rotational symmetry number of the  $n$  reactant extracted from its frequency output file.
- \* **Description:** Rotational symmetry number of the  $n$  reactant, where  $n$  can take the values 1 or 2 depending on the molecularity of the reaction.
- \* **Format:** NSYM\_R1 12

**NSYM\_RC**

- \* **Type:** Integer
- \* **Default value:** Rotational symmetry number of the pre-reactive complex extracted from its frequency output file.
- \* **Description:** Rotational symmetry number of the pre-reactive complex.
- \* **Format:** NSYM\_RC 6

**NSYM\_TS**

- \* **Type:** Integer
- \* **Default value:** Rotational symmetry number of the transition state extracted from its frequency output file.
- \* **Description:** Rotational symmetry number of the transition state.
- \* **Format:** NSYM\_TS 3

**NSYM\_PC**

- \* **Type:** Integer
- \* **Default value:** Rotational symmetry number of the product complex extracted from its frequency output file.
- \* **Description:** Rotational symmetry number of the product complex.
- \* **Format:** NSYM\_PC 1

**NSYM\_Pn**

- \* **Type:** Integer
- \* **Default value:** Rotational symmetry number of the  $n$  product extracted from its frequency output file.
- \* **Description:** Rotational symmetry number of the  $n$  product, where  $n$  can take the values 1 or 2.
- \* **Format:** NSYM\_P1 6

**RXSYM**

- \* **Type:** Integer
- \* **Default value:** 1
- \* **Description:** Reaction symmetry or degeneracy path.
- \* **Format:** RXSYM 4

**METHOD**

- \* **Type:** String
- \* **Default value:** TST, CVT
- \* **Description:** Kinetic method
- \* **Format:** METHOD TST

**TUNNEL**

- \* **Type:** Multiple choice



- \* **Options:** WIG, ECK
- \* **Description:** Tunneling
- \* **Format:** TUNNEL ECK

### **HINDROT**

- \* **Type:** String
- \* **Options:** PG, AS
- \* **Description:** Hindered rotor model
- \* **Format:** HINDROT PG

### **TEMP**

- \* **Type:** Float
- \* **Default value:** In the gas phase is 298.15 K. However, in solution it does not have a default value due to the solvent viscosity depends on the temperature.
- \* **Unit:** Kelvin
- \* **Description:** Temperature as a range (initial, final, and step) or list.
- \* **Format:** TEMP 100 500 10  
TEMP 100 200 300 400 500

### **STEMP**

- \* **Type:** Float
- \* **Description:** Special temperature as a list. Such temperature(s) is/are added to the values introduced by the keyword TEMP.
- \* **Format:** STEMP 298.15 316.4 456.15

### **PHASE**

- \* **Type:** Multiple choice
- \* **Default value:** GAS
- \* **Options:** GAS, SOL



- \* **Description:** Reaction phase
- \* **Format:** PHASE GAS

## **PATHPOINTS**

- \* **Type:** String
- \* **Description:** Directory containing the *Gaussian* output files, with the electronic energy and frequency analysis, of each snapshot along the IRC. It is mandatory for computing CVT rate constants. The outputs must be named according to the forward direction of the IRC (*i.e.*, from reactants to products) as 0001.out, 0002.out, 0003.out, etc. and stored in a directory within the current working directory (cwd). *Eyringpy* has a script to create a directory (called ircpoints) with the *Gaussian* input files of each point of the IRC. The code names the inputs as 0001.inp, 002.inp, 0003.inp, etc. To turn on this option only the following keys must be present in the \*.eif file: METHOD, IRC, REACT $n$ , PROD $n$ , and TS. Furthermore, METHOD must have the value CVT.
- \* **Format:** IRCPOINTS cwd/ircpoints

## **IRCPOINTS\_INP**

- \* **Type:** String
- \* **Description:** Set of keywords needed in the *Gaussian* input file(s) (\*.inp) to perform the frequency analysis of each snapshot of the IRC. Four values are required in the next order: memory, functional, basis, freq, and other optional keys (*e.g.*, Nosymm). It can be activated only if the IRC key is present. The inputs are named according to the forward direction of the IRC (*i.e.*, from reactants to products) as 0001.inp, 0002.inp, 0003.inp, etc. and stored in the directory cwd/nbo\_inp.
- \* **Format:** IRCPOINTS\_INP 8GB BHandHLYP aug-cc-pVDZ Freq Nosymm

## **RXCOORD**

- \* **Type:** Float
- \* **Default value:** Reaction coordinate of each snapshot extracted by *Eyringpy* from the



*Gaussian* output file with the electronic energy and frequency analysis. It is important to highlight *Eyringpy* parses this value from the IRC and writes it in the *Gaussian* input files \*.inp.

- \* **Units:** Bohr (mass-scaled) or  $\text{amu}^{1/2}$  Bohr (mass-weighted).
- \* **Description:** Reaction coordinate of each point along IRC for CVT. The value for TS ( $s = 0.0$ ) is introduced automatically by *Eyringpy*.
- \* **Format:** RXCOORD -0.3 -0.2 -0.1 0.1 0.2 0.3

### **TOR\_FREQ\_Rn**

- \* **Type:** Float
- \* **Default value:** Torsional frequencies of the  $n$  reactant extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:**  $\text{cm}^{-1}$
- \* **Description:** Torsional frequencies of the  $n$  reactant for the HR model.
- \* **Format:** TOR\_FREQ\_R1 89.2467 100.6895

### **TOR\_FREQ\_RC**

- \* **Type:** Float
- \* **Default value:** Torsional frequencies of the pre-reactive complex extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:**  $\text{cm}^{-1}$
- \* **Description:** Torsional frequencies of the pre-reactive complex for the HR model.
- \* **Format:** TOR\_FREQ\_RC 89.2467 100.6895

### **TOR\_FREQ\_TS**

- \* **Type:** Float
- \* **Default value:** Torsional frequencies of the TS extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:**  $\text{cm}^{-1}$





\* **Description:** Torsional frequencies of the TS for the HR model.

\* **Format:** TOR\_FREQ\_TS 89.2467 100.6895

### **TOR\_FREQ\_PC**

\* **Type:** Float

\* **Default value:** Torsional frequencies of the product complex extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.

\* **Units:**  $\text{cm}^{-1}$

\* **Description:** Torsional frequencies of the product complex for the HR model.

\* **Format:** TOR\_FREQ\_PC 89.2467 100.6895

### **TOR\_FREQ\_Pn**

\* **Type:** Float

\* **Default value:** Torsional frequencies of the  $n$  product extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.

\* **Units:**  $\text{cm}^{-1}$

\* **Description:** Torsional frequencies of the  $n$  product for the HR model.

\* **Format:** TOR\_FREQ\_P1 89.2467 100.6895

### **RED\_MOM\_INERT\_Rn**

\* **Default value:** Reduced moment(s) of inertia of the  $n$  reactant extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.

\* **Units:** amu Bohr<sup>2</sup>

\* **Description:** Reduced moment(s) of inertia of the  $n$  reactant for the HR model.

\* **Type:** Float

\* **Format:** RED\_MOM\_INERT\_R1 89.2467 100.6895

### **RED\_MOM\_INERT\_RC**

\* **Default value:** Reduced moment(s) of inertia of the pre-reactive complex extracted by



*Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.

- \* **Units:** amu Bohr<sup>2</sup>
- \* **Description:** Reduced moment(s) of inertia of the pre-reactive complex for the HR model.
- \* **Type:** Float
- \* **Format:** RED\_MOM\_INERT\_RC 89.2467 100.6895

### **RED\_MOM\_INERT\_TS**

- \* **Default value:** Reduced moment(s) of inertia of the TS extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:** amu Bohr<sup>2</sup>
- \* **Description:** Reduced moment(s) of inertia of the TS for the HR model.
- \* **Type:** Float
- \* **Format:** RED\_MOM\_INERT\_RC 89.2467 100.6895

### **RED\_MOM\_INERT\_PC**

- \* **Default value:** Reduced moment(s) of inertia of the product complex extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:** amu Bohr<sup>2</sup>
- \* **Description:** Reduced moment(s) of inertia of the product complex for the HR model.
- \* **Type:** Float
- \* **Format:** RED\_MOM\_INERT\_PC 89.2467 100.6895

### **RED\_MOM\_INERT\_Pn**

- \* **Default value:** Reduced moment(s) of inertia of the *n* product extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Units:** amu Bohr<sup>2</sup>
- \* **Description:** Reduced moment(s) of inertia of the *n* product for the HR model.
- \* **Type:** Float
- \* **Format:** RED\_MOM\_INERT\_R1 89.2467 100.6895

**TOR\_MIN\_Rn**

- \* **Type:** Integer
- \* **Default value:** Number of minima along the torsional coordinate for the  $n$  reactant extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Description:** Number of minima along the torsional coordinate for the  $n$  reactant for the HR model.
- \* **Format:** TOR\_MIN\_R1 1 2

**TOR\_MIN\_RC**

- \* **Type:** Integer
- \* **Default value:** Number of minima along the torsional coordinate for the pre-reactive complex extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Description:** Number of minima along the torsional coordinate for the pre-reactive complex for the HR model.
- \* **Format:** TOR\_MIN\_RC 1 3

**TOR\_MIN\_TS**

- \* **Type:** Integer
- \* **Default value:** Number of minima along the torsional coordinate for the TS extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Description:** Number of minima along the torsional coordinate for the TS for the HR model.
- \* **Format:** TOR\_MIN\_TS 2 3

**TOR\_MIN\_PC**

- \* **Type:** Integer
- \* **Default value:** Number of minima along the torsional coordinate for the product complex



extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.

- \* **Description:** Number of minima along the torsional coordinate for the product complex for the HR model.
- \* **Format:** TOR\_MIN\_PC 3 3

### **TOR\_MIN\_Pn**

- \* **Type:** Integer
- \* **Default value:** Number of minima along the torsional coordinate for the *n* product extracted by *Eyringpy* from the *Gaussian* output file with the frequency and HR analysis.
- \* **Description:** Number of minima along the torsional coordinate for the *n* product for the HR model.
- \* **Format:** TOR\_MIN\_P1 2 1

### **DIFF**

- \* **Type:** Bool
- \* **Description:** Diffusion effect
- \* **Format:** DIFF TRUE

### **AB**

- \* **Type:** Integer
- \* **Description:** Type of acid-base species. It is used only for reactions in water. The fully deprotonated species are denoted by 0, the monoanionic by 1, and the dianionic by 2.
- \* **Format:** AB 1

### **ET**

- \* **Type:** String
- \* **Description:** Marcus Theory for single electron transfer reactions.
- \* **Format:** ET MARCUS

**SOLVENT**

- \* **Type:** Multiple choice
- \* **Options:** WATER, PENTANE, HEXANE, HEPTANE, OCTANE
- \* **Description:** Solvent
- \* **Format:** SOLVENT WATER

**VISC**

- \* **Type:** Float
- \* **Description:** Solvent viscosity. *Eyringpy* has a database of viscosities (obtained from NIST: <https://webbook.nist.gov/chemistry/fluid/>) for several solvents, such as water, pentane, hexane, heptane, and octane, for temperatures at which they are in the liquid state. However, if the required solvent viscosity is not in our database, the user can supply it by the keyword VISC.  
Water: from 274 K to 373 K in 0.5 K steps.  
Pentane: from 144 K to 308 K in 0.5 K steps.  
Hexane: from 178 K to 340 K in 0.5 K steps.  
Heptane: from 183 K to 370 K in 0.5 K steps.  
Octane: from 217 K to 398 K in 0.5 K steps.
- \* **Unit:** Pa s
- \* **Format:** VISC 8.90E-4 8.54E-4 6.94E-4

**RXD**

- \* **Type:** Float
- \* **Description:** Reaction distance
- \* **Unit:** Ångström
- \* **Format:** RXD 2.6

**RADn**

- \* **Type:** Float



- \* **Description:** Radii of the solutes, where  $n$  can take the values 1 or 2 according to the molecularity of the reaction.
- \* **Unit:** Ångström
- \* **Format:** RAD1 2.59

### **PH**

- \* **Type:** Float
- \* **Default value:** 0 14 1
- \* **Description:** pH of water as range (initial, final, and step) or list.
- \* **Format:** PH 6 7 1  
PH 5 6 7 8

### **SPH**

- \* **Type:** Float
- \* **Description:** Special pH as a list. This pH(s) is/are added to the values introduced by the keyword PH.
- \* **Format:** SPH 7.4 10.5 13.2

### **PKA\_R1**

- \* **Type:** Float
- \* **Description:** pKa(s) of the reactant.
- \* **Format:** PKA\_R1 9.25

## 2.3 IRC-Analysis

### **IRC**

- \* **Type:** String
- \* **Description:** Frequency output filename(s) of the IRC. In the case of *Gaussian* (09 and 16), the forward and reverse directions of the IRC can be given in the same file or in



separate files (<https://gaussian.com/irc/>). *Eyringpy* accepts both formats.

- \* **Format:** IRC irc.out  
IRC irc\_for.out irc\_rev.out

### **RFORCE**

- \* **Type:** Bool
- \* **Description:** Reaction force analysis (RFA)
- \* **Format:** RFORCE TRUE

### **INTERP**

- \* **Type:** String
- \* **Default value:** SPLINE
- \* **Description:** Interpolation method for the RFA.
- \* **Format:** INTERP SPLINE

### **ORDER**

- \* **Type:** Integer
- \* **Default value:** 3
- \* **Description:** Spline interpolation order. It takes values between 3 and 5.
- \* **Format:** ORDER 5

### **NPOINTS**

- \* **Type:** Integer
- \* **Default value:** 0
- \* **Description:** Number of new points to generate for the spline interpolation. A thousand points produce smoother energy and reaction force profiles, but this can change the positions of the minimum and maximum of the reaction force.
- \* **Format:** NPOINTS 1000

**DISCRIM**

- \* **Type:** Bool
- \* **Description:** It removes some (or all) the points along the IRC causing noise in the energy profiles, conserving the values of the reaction and activation energies. It can be used only if the keyword `RFORCE` is used.
- \* **Format:** `DISCRIM TRUE`

**NBOINPUT**

- \* **Type:** String
- \* **Description:** It makes the input file (\*.inp) of *Gaussian* (09 or 16) to compute the NBO for each snapshot of the IRC. It requires four values in the next order: memory, functional, basis, and the value of the keyword POP (<https://gaussian.com/population>). It can be activated only if the IRC keyword is present. The inputs are named according to the forward direction of the IRC (*i.e.*, from reactants to products) as 0001.inp, 0002.inp, 0003.inp, etc. and stored in the `current_working_directory/nbo_inp`.
- \* **Format:** `NBOINPUT 10GB TPSSTPSS def2TZVP NBOREAD`

**NBODIRECTORY**

- \* **Type:** String
- \* **Description:** Directory containing the output files of *Gaussian* (09 or 16) (\*.out or \*.log) with the NBO data for each snapshot along the IRC. It is necessary for the population analysis (bond order, charge, dipole moment, and orbital energy). The outputs must be named according to the forward direction of the IRC (*i.e.* from reactants to products) as 0001.out, 0002.out, 0003.out, etc. and stored in the `current_working_directory/nbo_out`.
- \* **Format:** `NBODIRECTORY directoryname`

**HOMOLUMO**

- \* **Type:** Bool





- \* **Description:** HOMO-LUMO analysis. It can be used only if the NBODIRECTORY is supplied by the user.
- \* **Format:** HOMOLUMO TRUE

## **DIPOLE**

- \* **Type:** Bool
- \* **Description:** Dipole moment analysis. It can be used only if the NBODIRECTORY is supplied by the user.
- \* **Format:** DIPOLE TRUE

## **CHARGE**

- \* **Type:** Integer
- \* **Description:** Atom(s) (*i.e.*, their number(s)) whose charge along the IRC is required to monitor. Atoms must be separated by a blank space. It must be present when the natural charge analysis is invoked. It can be used only if the NBODIRECTORY is supplied by the user.
- \* **Format:** CHARGE 1 5 9 10

## **BONDLENGTH**

- \* **Type:** Integer
- \* **Description:** Atoms (*i.e.*, their numbers) forming the bond(s) to be monitored along the IRC. The atoms of each bond must be separated by a hyphen (-) and each bond must be separated by a blank space. It is mandatory if the bond length analysis is needed.
- \* **Format:** BONDLENGTH 1-2 3-4 1-4

## **WIBERG**

- \* **Type:** Integer
- \* **Description:** Atoms (*i.e.*, their numbers) forming the bond(s) for which is required to analyze the Wiberg bond index evolution along the IRC. The atoms of each bond must be



separated by a hyphen (-) and each bond must be separated by a blank space. It must be present if the analysis of the Wiberg bond index is needed. It can be used only if the NBODIRECTORY is supplied by the user.

- \* **Format:** WIBERG 1-2 3-4 1-4

## ANGLE

- \* **Type:** Integer
- \* **Description:** Atoms (*i.e.*, their numbers) forming the angle(s) to be monitored along the IRC. The atoms of each angle must be separated by a hyphen (-) and each angle must be separated by a blank space. It is mandatory if the angle analysis is needed.
- \* **Format:** ANGLE 1-2-3 2-3-4

## XYZ

- \* **Type:** Bool
- \* **Description:** It creates \*.xyz file with the geometries of the snapshots along the IRC.
- \* **Format:** XYZ TRUE

## PLOT

- \* **Type:** Multiple choice
- \* **Options:** RFORCE, DIPOLE, HOMOLUMO, CHARGE, BONDLENGTH, WIBERG, ANGLE.
- \* **Description:** Property(ies) monitored along the IRC for which the corresponding results are required to plot.
- \* **Format:** PLOT RFORCE HOMOLUMO WIBERG ANGLE



## 3 The *Eyringpy* output file

Here we describe the terms used in the different types of *Eyringpy* output files (\*.eof). The general *Eyringpy* output file is named as the directory in which the computation of interest is performed (*i.e.*, foldername.eof). However, for the case of the analysis of primitive changes along the IRC there are other output files whose names are given in the foldername.eof file.

### 3.1 Thermochemistry

- \* **DU**: Reaction internal energy in kcal mol<sup>-1</sup>.
- \* **DS**: Reaction entropy in kcal mol<sup>-1</sup> K<sup>-1</sup>.
- \* **DH**: Reaction enthalpy in kcal mol<sup>-1</sup>.
- \* **sym**: The symmetry numbers of the species participating in the reaction are included in the partition functions.
- \* **nosym**: The symmetry numbers of the species participating in the reaction are not included in the partition functions.
- \* **latm**: Reaction Gibbs free energy in terms of partial pressures (1 atm).
- \* **1M**: Reaction Gibbs free energy in terms of concentrations (1 M).
- \* **DHf++**: Activation enthalpy at 0 K for the forward direction in kcal mol<sup>-1</sup>.
- \* **DHr++**: Activation enthalpy at 0 K for the reverse direction in kcal mol<sup>-1</sup>.
- \* **DU++**: Activation internal energy in kcal mol<sup>-1</sup>.
- \* **DS++**: Activation entropy in kcal mol<sup>-1</sup> K<sup>-1</sup>.
- \* **DH++**: Activation enthalpy (for the forward direction) in kcal mol<sup>-1</sup>.
- \* **DE**: Reaction electronic energy for SET reactions in kcal mol<sup>-1</sup>.
- \* **DG++**: Gibbs activation energy for SET reactions in kcal mol<sup>-1</sup>.
- \* **Lambda**: Reorganization energy for SET reactions in kcal mol<sup>-1</sup>.
- \* **DEfor++**: Activation electronic energy for the forward direction in kcal mol<sup>-1</sup>.
- \* **DErev++**: Activation electronic energy for the reverse direction in kcal mol<sup>-1</sup>.



- \* **DERxn**: Reaction electronic energy in kcal mol<sup>-1</sup>.

### 3.2 Kinetics

#### Rate constants

- \* **k**: Rate constants without the reaction symmetry.
- \* **ksym**: Rate constants including the reaction symmetry.
- \* **kwig**: Rate constants including the Wigner tunneling.
- \* **keck**: Rate constants including the Eckart tunneling.
- \* **kapp**: Rate constants including the diffusion effect.
- \* **kf**: Rate constants including the pH effect.

**Note:** The units of the rate constants for unimolecular reactions are s<sup>-1</sup> (in the gas phase and in solution), cm<sup>3</sup> molecule<sup>-1</sup> s<sup>-1</sup> for bimolecular reactions in the gas phase and M<sup>-1</sup> s<sup>-1</sup> for bimolecular reactions in solution.

- \* **RxCoord**: Reaction coordinate.

#### Tunneling

- \* **wig**: Wigner transmission coefficients.
- \* **eck**: Eckart transmission coefficients.

#### Diffusion

- \* **visc**: Solvent viscosity in Pa s.
- \* **kd**: Diffusion constant in M<sup>-1</sup> s<sup>-1</sup>.

#### pH effect

- \* **f0**: Molar fraction of the fully deprotonated specie (*e.g.*, NH<sub>3</sub>).
- \* **f1**: Molar fraction of the mono-ion (*e.g.*, NH<sub>4</sub><sup>+</sup>).

### 3.3 IRC-Analysis

#### Reaction Force

- \* **Reactants**: Reaction coordinate of the reactant(s).
- \* **Force minimum**: Reaction coordinate of the force minimum.



- \* **Transition state:** Reaction coordinate of the transition state.
  - \* **Force maximum:** Reaction coordinate of the force maximum.
  - \* **Products:** Reaction coordinate of the product(s).
  - \* **Wfor+++:** Activation work for the forward direction.
  - \* **Wrev++:** Activation work for the reverse direction.
  - \* **W1:** Work of the reactant region.
  - \* **W2:** Work of the transition state region.
  - \* **W3:** Work of the transition state region.
  - \* **W4:** Work of the product region.
  - \* **Wrxn:** Reaction work.
  - \* **RxCoord:** Reaction coordinate ( $\xi$ ).
  - \* **Energy:** Electronic energy in kcal mol<sup>-1</sup>.
  - \* **Force:** Reaction force in kcal mol<sup>-1</sup>  $\xi^{-1}$
- Note:** The units of the reaction coordinate ( $\xi$ ) depend on the coordinates (cartesian or mass-weighted) of the IRC computation. Work in kcal mol<sup>-1</sup>.

#### Property analysis

- \* **Distance:** Bond length in Ångström.
- \* **Angle:** Angle in Degrees.
- \* **Wiberg:** Wiberg bond index
- \* **Charge:** Charge in |e|.
- \* **Dipole moment:** Dipole moment in Debye.
- \* **Homo-Lumo:** Homo-Lumo in eV.

## 4 References

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